https://doi.org/10.1590/S1413-4152031S104

Scientific Paper

Microaeration of Upflow Anaerobic Sludge Blanket reactor using desorption chamber gas: methane recovery and H2S removal

Daniel Carvalho Mota[1], Isabela Ferreira Batista[1], Cláudio Leite de Souza[2]*

ABSTRACT

This study evaluated the integration between two technologies, a demonstration-scale desorption chamber (DC) and microaeration, in a demonstration-scale UASB reactor treating real domestic sewage, aiming to enhance methane recovery and reduce hydrogen sulfide. The DC removes dissolved CH_4 and H_2S from liquid, generating a waste gas (containing CH_4, CO_2, H_2S, and air - O_2 and N_2), which was used in this study to microaerate the UASB reactor. The DC achieved median removal efficiencies of 55% for dissolved methane and 79% for dissolved hydrogen sulfide, despite the low gas-to-liquid ratio applied (rQ ≈ 0.042). Waste gas extracted from the DC was successfully reused as the microaeration source in the UASB reactor, resulting in a median 28% reduction in H_2S mass flow and a 60% decrease in biogas H_2S concentration, while also increasing methane mass flow by approximately 37%, even though the doses of O_2 and operational conditions varied substantially throughout the study. The integration of a desorption chamber and a microaerated UASB reactor produced biogas with a typical composition of ~50–54% CH_4 and ~800 ppm_v H_2S, suitable for thermal energy recovery without any additional treatment or reducing the required treatment for energy recovery (biogas generator). The demonstration-scale conditions highlighted both the technical feasibility of the approach and the operational challenges of controlling microaeration under real wastewater variability, indicating a need for further research to standardize dosing strategies (L O_2/L biogas instead of L O_2/L effluent) to support full-scale implementation.

Keywords: waste gas; biogas; energy recovery; odor nuisance; sulfide oxidation.

INTRODUCTION

Anaerobic reactors, particularly the Upflow Anaerobic Sludge Blanket (UASB) type, are widely used for sewage treatment in Brazil due to their operational simplicity, low sludge production, low energy consumption, and biogas generation, among other advantages. However, they also present limitations, including the need for post-treatment and challenges in managing biogas and dissolved gases, particularly methane (CH_4), a potent greenhouse gas, and hydrogen sulfide (H_2S), which is associated with odor, corrosion, and toxicity (Bressani-Ribeiro and Chernicharo, 2024).

Dissolved methane (CH_4-D) has become a central topic in discussions regarding greenhouse gas emissions from wastewater treatment centers, especially in anaerobic systems. Unlike the methane recovered in the gaseous phase, this fraction remains in the liquid effluent. It may be gradually released into the atmosphere through desorption during transport or in downstream units within the WWTP. This release can increase overall emissions, as CH_4-D may account for 40% of the methane produced during treatment (Souza, Chernicharo and Aquino, 2011; Souza, Chernicharo and Melo, 2012). According to the IPCC (2019), methane emissions in anaerobic reactors are typically estimated based on the removed organic load, the maximum methane production potential (Bo), and the methane conversion factor (MCF), assuming that all methane is emitted as biogas. In this context, unaccounted CH_4-D may represent a significant source of fugitive emissions, especially when no removal, capture, or recovery strategies are in place, potentially affecting the facility's GHG balance and decreasing the system's energy efficiency.

Hydrogen sulfide (H_2S) is also a particularly problematic dissolved gas., A colorless, toxic, and corrosive compound, denser than air and partially soluble in water, it is characterized by a "rotten egg" odor and detectable by the human olfactory system at extremely low concentrations, on the order of 0.00047 ppm_v (Metcalf and Eddy, 2016). As a result, it is a major contributor to operational issues in wastewater treatment plants, including odor nuisance, corrosion of

[1]Universidade Federal de Minas Gerais, Department of Sanitation, Environment and Water Resources – Belo Horizonte (MG), Brazil.
[2]Universidade Federal de Minas Gerais, Department of Sanitary and Environmental Engineering – Belo Horizonte (MG), Brazil.
***Corresponding author:** claudio@desa.ufmg.br
Conflicts of interest: the authors declare no conflicts of interest.
Funding: Conselho Nacional de Desenvolvimento Científico e Tecnológico – CNPq (Grant number 421289/2023-6); Fundação de Amparo à Pesquisa do Estado de Minas Gerais – FAPEMIG (Grant number APQ-03082-26; APQ-06009-26); Agência Reguladora de Saneamento e Energia de Minas Gerais – ARSAE-MG, Companhia de Saneamento de Minas Gerais – COPASA, Aliança Estratégica UFMG/COPASA (Grant number 22.3157).
Received on: 12/25/2025 – **Accepted on:** 03/11/2026
Editor: André Bezerra dos Santos, Universidade Federal do Ceará, Fortaleza, Ceará/CE, Brasil.

concrete and metallic structures, damage to biogas energy recovery systems, occupational health risks, and the formation of sulfur oxides during gas combustion, which may contribute to acid rain. In anaerobic reactors, H_2S is predominantly present in the dissolved phase, accounting for 75.1% of the sulfide generated and corresponding to 57–64% of the influent sulfur. In contrast, the fraction present in the biogas is comparatively small, representing only 1.3% of the hydrogen sulfide formed and 0.9% of the influent sulfur (Souza, 2010; Souza, Chernicharo and Melo, 2012).

One technique used to mitigate these issues is the desorption chamber (DC), which removes dissolved gases from anaerobic effluents (Glória *et al.*, 2016; Huete *et al.*, 2018). In this process, a large mass of CH_4 and H_2S is desorbed from the liquid phase and/or partially oxidized, generating a secondary gas stream, referred to as waste gas, which also contains significant amounts of air (N_2 and O_2).

In this study, this waste gas, typically a residual stream requiring treatment, was repurposed as a source of microaeration for the UASB reactor, reintroducing the dissolved gases removed from the system. Microaeration, defined as the controlled introduction of small amounts of oxygen, has emerged as a promising strategy to enhance anaerobic reactor performance (Castro, Azevedo and Souza, 2021; Fu *et al.*, 2023), as it can improve methane yield, reduce H_2S concentrations, and stabilize the process. However, challenges such as precise oxygen dosing, lack of standardization, and limited operational guidance still hinder large-scale implementation (Fu *et al.*, 2023). Due to these challenges and to support the interpretation of some results of this study, one goal is to explore a more stable and efficiency-based parameter for microaeration.

Another aspect is that the research on microaeration has largely focused on sludge and high-strength waste digesters, mainly at a laboratory or pilot scale. At the same time, few studies have explored its application to domestic sewage, particularly in full-scale UASB systems (Chen *et al.*, 2020). In this context, the current study evaluated the integration of a desorption chamber with microaeration by applying the waste gas stream in a UASB reactor treating real domestic sewage, aiming to improve energy recovery, enhance system performance, and environmental outcomes.

METHOD

The experimental setup consisted of a demonstration-scale Upflow Anaerobic Sludge Blanket (UASB) reactor, a desorption chamber (DC) filled with support media (3-inch Pall rings), a peristaltic pump, and tubular microbubble diffusers. The main parameters were listed in Table 1 as well as the dimensions of the system. Figure 1 illustrates a system sketch (a) and the real units (b).

Table 1 – Dimensions of the UASB reactor and the Desorption Chamber (DC).

Characteristic	UASB Reactor	Desorption Chamber
Diameter (m)	2.00	0.35
Height (m)	4.50	1.20
Effective Volume (m^3)	14.00	0.12
HRT[1] (h)	9.5	-
Material	Ferrocement (reactor body); Glass fiber (three-phase separator)	Glass fiber; 1.00 m of support media consisting of 3-inch plastic Pall rings

[1]HRT – Hydraulic Retention Time: theoretical time required for a fluid to pass through a given volume.

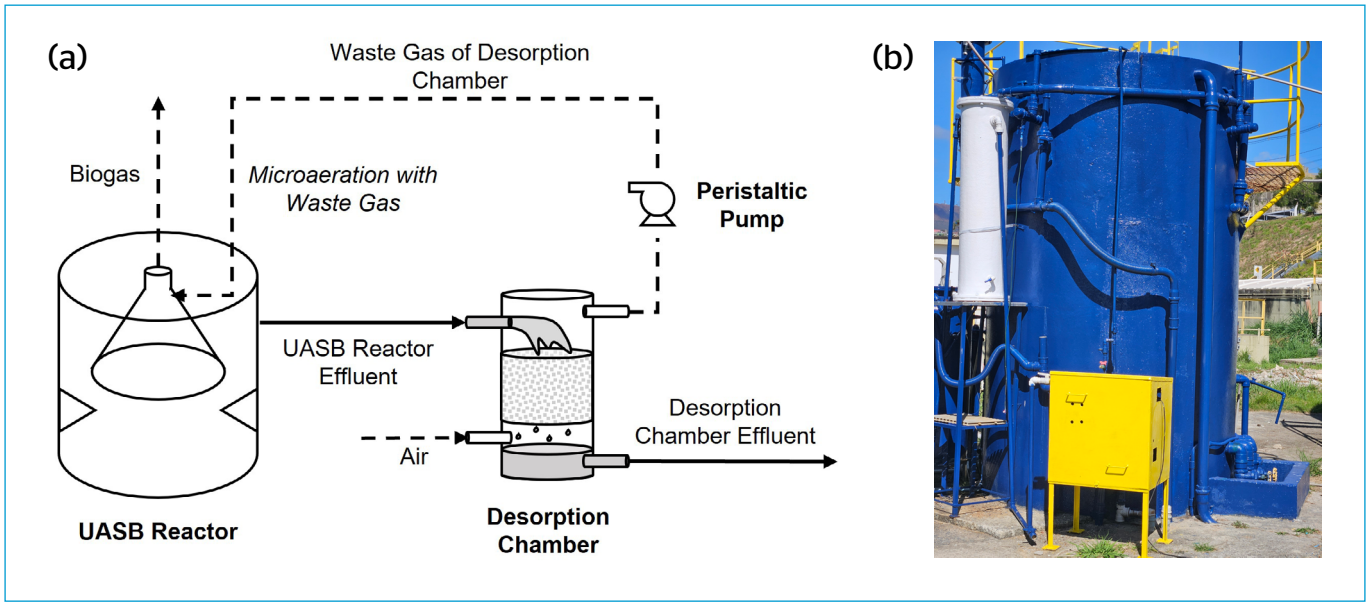


Figure 1 – a) Schematic sketch of the microaerated UASB reactor system with waste gas from the desorption chamber (DC); b) UASB reactor (dark blue) and desorption chamber (white) used in the research.

The UASB reactor, which had been in operation for ten years prior to the start of the experiment, was fed with approximately 1.48 $m^3\ h^{-1}$ of domestic wastewater, resulting in a hydraulic retention time (HRT) of approximately 9.5 h and volumetric organic loading rate of 0.96 ± 0.55 kg COD $m^{-3}\ day^{-1}$. The desorption chamber received the entire effluent from the UASB reactor, ensuring full system integration. In this unit, the average hydraulic loading rate (HLR) was 0.26 ± 0.04 $m^3\ m^{-2}\ min^{-1}$, corresponding to applied flow rates of 0.41 ± 0.05 L s^{-1}. Gas extraction from the desorption chamber was performed using a Masterflex peristaltic pump with variable speed control (0–600 rpm).

To evaluate the different microaeration application points and compare them with the conventional UASB reactor, three main phases were established. The study began with a control anaerobic phase (F0), during which the desorption chamber was monitored and adjusted for the subsequent use of waste gas for microaeration. This phase was characterized by low gas flow rates and, consequently, low gas-to-effluent flow ratios (rQ), with an average value of 0.042 ± 0.010. After this initial period, Phase F0 was applied at shorter intervals between transitions or whenever microaeration was interrupted due to unit maintenance (Figure 2). Consequently, the control anaerobic phase, later used as a reference for comparison with the microaerated reactor, was distributed chronologically throughout the entire experiment, resulting in mean values that were more representative and less influenced by seasonal factors (rainy periods, temperature variations, and water consumption habits).

Subsequently, Phases 1 and 2 were implemented. These experimental phases involved evaluating different microaeration injection points within the UASB reactor. Two points of application were tested: in the headspace of the three-phase separator, approximately in the middle, and submerged in the water column of the three-phase separator, at a level between 0.5 and 1,0 m below the water surface. The experiment was divided into three experimental phases, as shown in Table 2. The desorption chamber observation followed the same microaeration phases to simplify analysis and comparisons. The experiment timeline can be found in Figure 2, focusing on the phase-change period.

System performance monitoring encompassed a range of physical-chemical and gas-related parameters, using standard methods and instrumentation. Influent and effluent flow rates were measured directly in the field. Dissolved methane concentrations were determined using the headspace extraction method followed by gas chromatography, as described by Souza, Chernicharo and Aquino (2011). Dissolved hydrogen sulfide was quantified by the colorimetric method, according to Plas *et al.* (1992) and pH, temperature, and dissolved oxygen were monitored with a multiparameter probe (HACH® HQ40D). Biogas production was quantified using a gas meter (Ritter® TG3), while biogas composition required a portable gas analyzer equipped with electrochemical sensors (LANDTEC® GEM™ 5000). Statistical analyses were conducted using the non-parametric Mann–Whitney U test, with a 5% significance level.

For the calculation of mass flow, flow rate data were normalized. A calculation of the biogas generated in the UASB reactor was carried out to estimate hydrogen sulfide removal efficiency and the increase in methane mass flow in the microaerated UASB reactor. The proper flow of biogas was calculated by subtracting the waste gas injected into the reactor from the total biogas with microaeration, while the composition was estimated from the average biogas composition obtained during the control anaerobic phase (F0).

After the initial data analysis, an additional evaluation was conducted to better correlate the microaeration dosage with the observed process efficiencies. Due to the high variability observed in the data, both in terms of H_2S removal

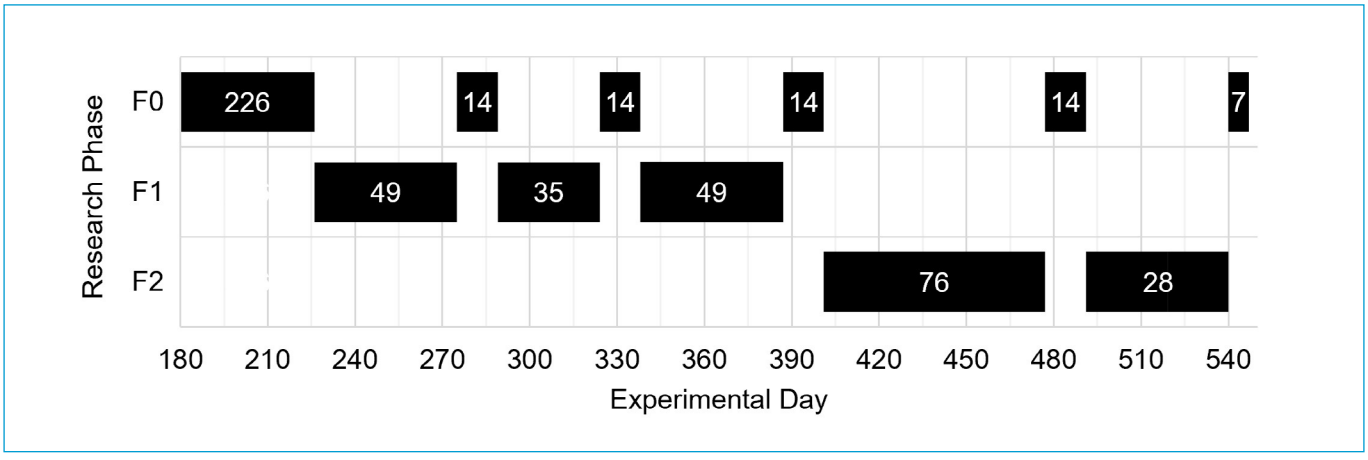


Figure 2 – Experiment timeline.

Table 2 – Operational parameters of the system in each experimental phase.

Experimental phase	Desorption Chamber			UASB microaeration	
	Influent Flow (L.s-1)	rQ (Qg. Ql^{-1})	O_2 – waste gas (% v/v)	Injection Point	Microaeration Rate (L O_2. L^{-1} influent)
Phase 0 (F0) – Anaerobic – Control	0.41 ± 0.07	0.042 ± 0.018	9.1 ± 1.9	–	–
Phase 1 (F1)	0.42 ± 0.03	0.043 ± 0.015	11.0 ± 1.5	Headspace	0.0051 ± 0.0009
Phase 2 (F2)	0.40 ± 0.02	0.042 ± 0.010	9.7 ± 1.6	Submerged	0.0050 ± 0.0010

efficiency and methane mass flow, alternative dosing parameters were investigated to identify a more suitable performance indicator.

In this context, different dosage parameters were evaluated based on previous studies (Krayzelova *et al.*, 2015) and the previous experience of the researchers. The analyses were performed using Microsoft Excel® (Microsoft 365) and the tested dosing parameters were:

1. the conventional microaeration rate expressed as liters of oxygen per liter of influent ($LO_2.L_{inf}^{-1}$),
2. liters of oxygen per liter of total biogas output (biogas produced plus waste gas used for microaeration) ($LO_2.L_{total\ biogas}^{-1}$), and
3. liters of oxygen per liter of biogas produced in the UASB reactor ($LO_2.L_{prod.\ biogas}^{-1}$).

Scatter plots with logarithmic trendlines were used to evaluate the most suitable dosing parameters. The analysis considered the coefficient of determination (R^2) obtained for each relationship.

RESULTS AND DISCUSSION

Desorption Chamber Performance

The experimental period in the desorption chamber lasted approximately 540 days, comprising phases with and without microaeration integration. Figure 3 presents the main results, with vertical dashed lines indicating the onset of microaeration (phases 1 and 2). The median removal efficiencies of dissolved methane and hydrogen sulfide were 55% and 79%, respectively, with maximum values of 78% and 99%, respectively.

These medians were lower than those reported in previous studies: Glória *et al.* (2016) achieved a 60% methane and 84% hydrogen sulfide removal at a gas-to-liquid ratio (rQ) of 3.1, Huete *et al.* (2017) achieved methane removal efficiencies of up to 99% with rQ values between 0.5 and 1.0, while Santo *et al.* (2019) reported 89% and 86% removals, respectively, with an rQ of 0.1. In contrast, Santo *et al.* (2019) also used a rQ of 0.05 in a very similar desorption chamber, achieving 43% methane and 54% hydrogen sulfide removal, indicating that rQ plays an important role in removal effectiveness.

The relatively lower efficiencies in the current study can be attributed to a group of aspects such as:

(1) the low average rQ applied (0.042), less than half the value used by Santo *et al.* (2019), which limited gas renewal at the interfaces and, consequently, mass transfer;
(2) the variability of the real domestic wastewater;
(3) higher hydraulic loading rate compared to Glória *et al.* (2016).

As illustrated in Figure 3b, the removal efficiencies of dissolved CH_4 and H_2S (%) and the methane concentration in the desorption chamber gas (% v/v) fluctuated throughout the experimental period. Regarding the methane concentration, values generally range from 20 to 30%.

Despite a few operational changes in the desorption chamber, establishing experimental phases was crucial to verify external influences and the relation between the DC and microaeration results. When the experimental phases are analyzed individually, clearer trends can be observed in the removal efficiencies of dissolved methane (CH_4) and dissolved hydrogen sulfide (H_2S) across the experimental phases (F0, F1, and F2), as presented in Figure 4. For CH_4 (Figure 4a), removal efficiencies predominantly ranged from 50 to 60%, with medians around 54 to 59% for all phases. Phases F0 and F1 exhibited lower data dispersion, indicating a more stable system behavior. In contrast, phase F2 showed greater variability, with minimum values near 20%, suggesting operational instability or a stronger influence of environmental factors on process performance.

Regarding H_2S (Figure 4b), removal efficiencies were higher than those observed for CH_4, with medians of 75–80%. However, dispersion was more

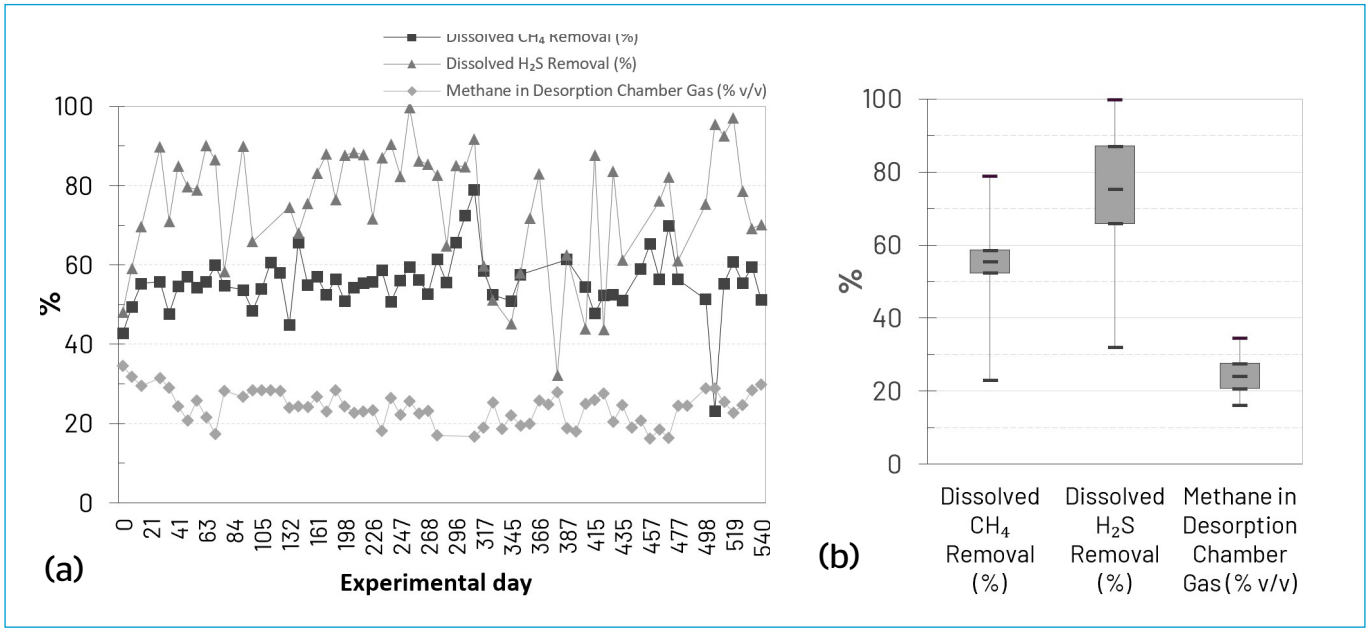


Figure 3 – a) Removal of dissolved CH_4 and H_2S (%) and methane concentration in the desorption chamber gas (% v/v) over the experimental days; b) Box plot of the variables presented.

pronounced, especially in phase F1, where values ranged from 30 to 100%, approximately. In phases F0 and F2, although the minimum values were slightly higher, significant variability persisted, indicating that H_2S removal efficiency is more sensitive to operational conditions and system fluctuations.

Figure 5 illustrates the concentrations of CH_4 (a) and H_2S (b) in the desorption chamber gas across the experimental phases F0, F1, and F2. For CH_4, median concentrations remained relatively stable across phases, ranging from approximately 21 to 26% v/v. Phase F2 exhibited the widest interquartile range, indicating greater variability in CH_4 concentration for the chamber during this phase. However, the minimum values were around 16–17% v/v across all phases.

For H_2S, phase F0 presented the lowest concentrations, with a median value of around 190 ppm_v and a relatively narrow dispersion, indicating more stable conditions. In contrast, Phase F1 exhibited a substantial increase in H_2S levels, with a median concentration close to 500 ppm_v and markedly higher variability, including maximum values near 1800 ppm_v. Phase F2 showed intermediate behavior, with a median concentration of 380 ppm_v and reduced dispersion compared to F1. Overall, the results indicate that H_2S concentrations in the waste gas were the lowest during F0 and most variable during F1, while F2 represented a partial recovery in stability.

*This behavior was also observed for methane during F1, confirming that this phase experienced greater operational and environmental instability. However, the median values were more comparable, particularly for F1 and F2, at approximately 500 and 350 ppm, respectively.*Figure 6a shows the mass flow rate of CH_4 in the waste gas across the three experimental phases. The median values remained

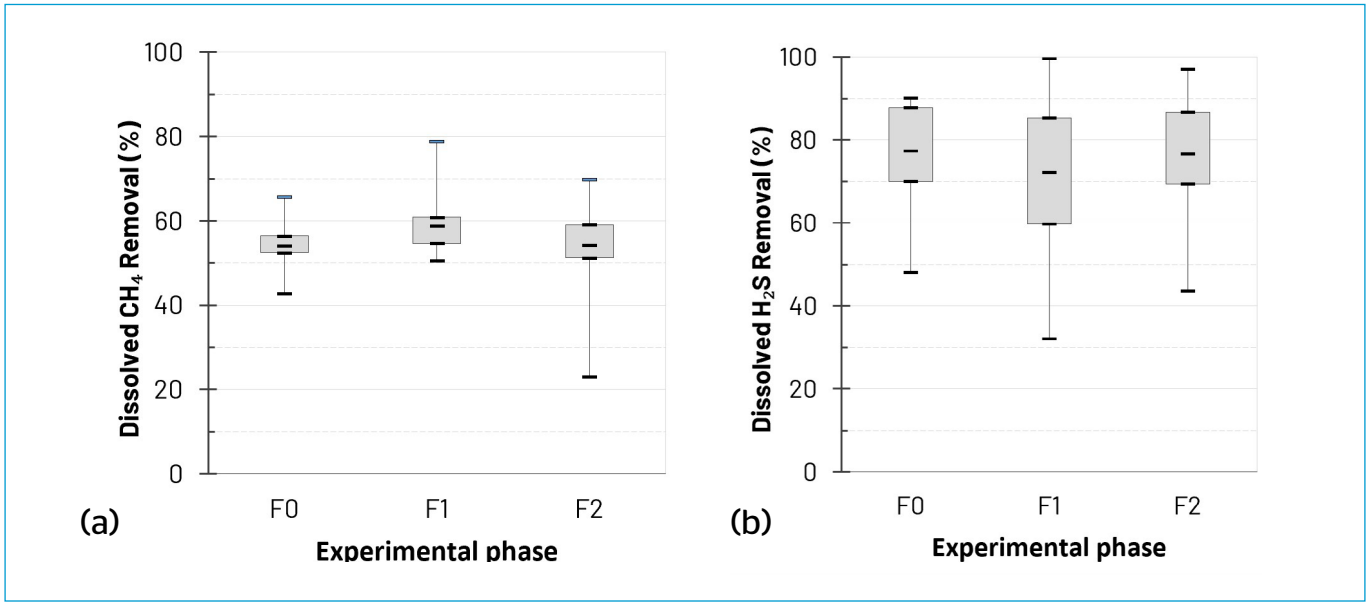


Figure 4 – Dissolved Gases Removal in Desorption Chamber.

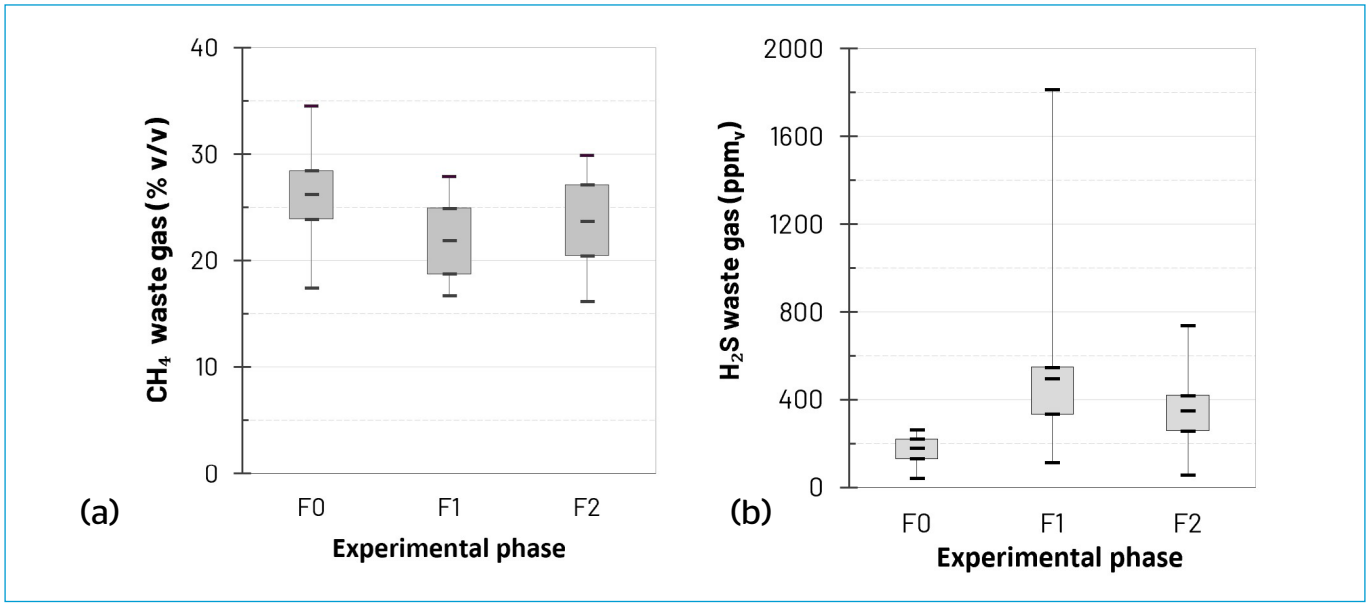


Figure 5 – Waste gas concentrations in Desorption Chamber, CH_4 (a) and H_2S (b).

relatively stable, at around 200 g/d. The interquartile range and whiskers indicate high variability throughout the experimental period. In contrast, the mass flow rate of H_2S (Figure 6b) exhibited more pronounced variations, indicating more susceptibility due to operational and environmental changes. In Phase F0 (control), the H_2S mass flow rate remained at low levels, with a median value around 0.3 g/d and limited variability. In Phase F1, there was a marked increase in H_2S mass flow, with a median close to 1.0 g/d and higher dispersion. It is understood that the higher concentration of oxygen in the gas phase of DC at this phase (11%) may have created more difficulties for the partial oxidation of the sulfide to elemental sulfur, which requires lower oxygen concentrations in the liquid phase, than it promoted the complete oxidation of sulfide to sulfate, thus increasing the possibility of desorption to the gas phase. In Phase F2, the H_2S mass flow decreased compared to F1, with a median of approximately 0.7 g/d and reduced variability.

Performance of UASB Microaeration with Waste Gas

The results presented in this section were obtained through the integration of DC waste gas and microaeration into the three-phase separator of the UASB reactor (phases 1 and 2, respectively). Figure 7a shows the H_2S mass flow (g d^{-1}) removal efficiencies, which exhibited high variability in both phases. In terms of median values, both removal phases were very similar, at approximately 28%. These median efficiencies did not differ significantly (5% significance level, Mann–Whitney nonparametric test). Therefore, although the microaeration injection points were different, headspace (F1) and submerged (F2), no clear

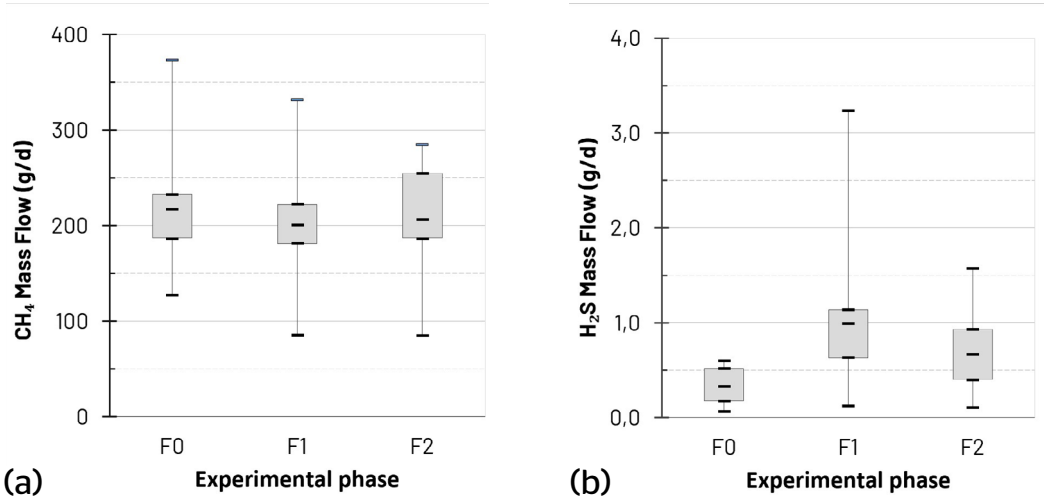


Figure 6 – Mass Flow of CH_4 (a) and H_2S (b) in the waste gas.

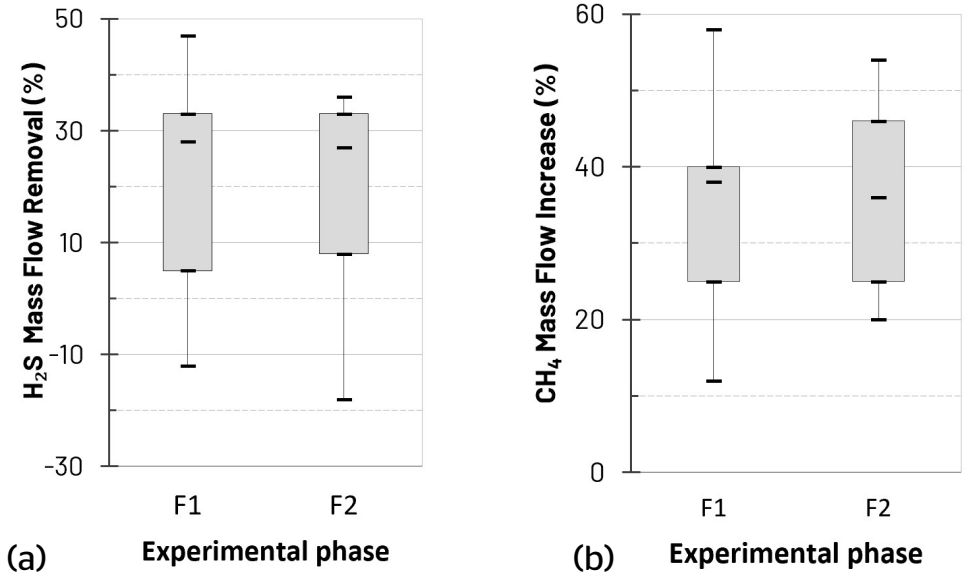


Figure 7 – Performance indicators of UASB microaeration in the experimental phases (F1 and F2): (a) H_2S mass flow removal (%); (b) CH_4 mass flow increase (%).

advantage was observed for the submerged application, despite the hypothesis that it would promote greater oxygen contact and direct microaeration in the liquid and scum layers.

The removal efficiencies obtained were lower than those reported in other studies. In review studies of air-based microaeration, Krayzelova *et al.* (2015) reported removal efficiencies consistently above 72%, and Chen *et al.* (2020) observed a minimum efficiency of 52% in diverse anaerobic reactors. Castro, Azevedo and Souza (2021) concluded that air addition in the digestion zone of a UASB reactor treating domestic sewage showed hydrogen sulfide removal efficiencies from biogas greater than 90%, and Kraakman *et at.* (2023) reported removal efficiencies of at least 80% for H_2S in large-scale, air microaerated wastewater municipal sludge digester biogas.

An important factor contributing to this lower efficiency is the gas used in microaeration, which has a lower oxygen concentration than air (approximately 10%). In addition, the waste gas from the DC presents median hydrogen sulfide concentrations of around 190 ppm_v (F0), 500 ppm_v (F1), and 380 ppm_v (F2), which also contribute to the H_2S content in the final biogas. Possibly the most important factor behind the understanding of the removal of H_2S around 30% is the residence time of the biogas in the headspace. This number accounted for a relatively low value of about 0.40 h, on average. Krayzelova *et al.* (2015), based on other studies, reported removal efficiencies that vary with residence time in the headspace, with removals of around 90% for residence times of 3–5 h and about 40% for residence times of approximately 1 h. However, achieving such high residence times in typical UASB reactors could result in substantial changes to the design parameters of the three-phase separators and an increase in associated costs.

On the other hand, the approximately 28% efficiency represents an interesting, cost-effective removal, although relatively low, as the resulting biogas presented median concentrations of 820 ppm_v. Compared to the typical H_2S concentration in biogas (2000 ppm_v), the concentration reduction was achieved around 60%. Such a concentration (820 ppm_v) directly meets the quality requirements for thermal energy recovery in boilers, without the need for additional treatment (< 1000 ppm_v; Wellinger and Lindberg, 2001).

As seen in Figure 7b, the relative increase in CH_4 mass flow (g d^{-1}) in the biogas was similar across operational phases F1 and F2, with a median of approximately 37% and considerable variability in the data. Statistical analysis indicated no significant differences between the two phases at the 5% significance level (Mann–Whitney test). Therefore, submerging the injection point for the DC waste gas did not provide an observable advantage in methane recovery.

The increase in methane mass flow was estimated through a mass balance approach. The total CH_4 mass flow measured in the mixed gas stream was compared with the expected CH_4 mass flow assuming that no DC waste gas was introduced. This reference condition was calculated considering the normalized total volumetric flow minus the normalized volumetric flow of waste gas, and assuming that the CH_4 concentration in the resulting flow would correspond to the average CH_4 concentration measured in the anaerobic phase.

Based on this calculation procedure, the observed increase in CH_4 mass flow is likely associated with the recovery of dissolved methane by the DC unit. In contrast, the possibility that microaeration occurring in the upper region of the three-phase separator significantly enhanced anaerobic biodegradation is considered low (Castro, Azevedo and Souza, 2021). Previous studies have shown that dissolved methane losses in the effluent of UASB reactors may reach 60% of the methane usually captured in the biogas (Souza, Chernicharo and Aquino, 2011).

From an energy recovery perspective, the 37% increase represents a potentially valuable gain in methane. Although some dilution of methane occurred due to mixing between biogas and DC waste gas, the resulting CH_4 concentration was around 54%, indicating that the dilution was not pronounced. The waste gas stream still contained relevant concentrations of recovered CH_4 and CO_2, approximately 22 and 5%, respectively. Moreover, methane concentrations around 54% remain suitable for most thermal and electrical energy conversion systems operating with biogas (Sun *et al.*, 2015; Wellinger and Lindberg, 2001).

Concerning the point of oxygen/gas application, submerged microaeration presented disadvantages compared to headspace injection. These included a trend toward lower removal efficiency, likely due to oxygen consumption in the liquid phase, as well as operational challenges (e.g., diffuser inspection and replacement) and increased system complexity. Supporting the preference for headspace application, most sulfur-oxidizing bacteria (SOB) are predominantly found in the headspace, indicating that sulfide oxidation is primarily occurring there (Krayzelova *et al.*, 2015).

Study for a more suitable parameter for microaeration dosage

Due to the high variability observed in the data, both in H_2S removal efficiency and in methane mass flow, alternative parameters were evaluated to better describe the relationship between microaeration dosage and process performance.

The coefficients of determination (R^2) obtained between the evaluated microaeration parameters and the process responses (hydrogen sulfide removal and methane mass flow increase) are summarized as follows:

(1) liters of oxygen per liter of influent ($LO_2.L_{inf}^{-1}$) resulted in R^2 values equal to zero for both variables;

(2) liters of oxygen per liter of total biogas output ($LO_2.L_{total\ biogas}^{-1}$) resulted in R^2 values of 0.46 for hydrogen sulfide removal and 0.23 for methane mass flow increase; and

(3) liters of oxygen per liter of biogas produced in the UASB reactor ($LO_2.L_{prod.\ biogas}^{-1}$) resulted in R^2 values of 0.60 and 0.41 for hydrogen sulfide removal and methane mass flow increase, respectively.

Therefore, the microaeration rate expressed relative to the biogas produced in the UASB reactor ($LO_2.\ L_{prod.\ biogas}^{-1}$) showed the strongest correlations with the evaluated performance indicators. This parameter excludes the volumetric contribution of the DC gas used for microaeration, thereby better representing the operational conditions of the reactor.

Considering this newly defined parameter, the microaeration rates obtained for operational phases F1 and F2 (Figure 8) were compared. No statistically significant difference was observed between the two phases (5% significance level, Mann–Whitney test). Therefore, data from both phases were grouped for the subsequent correlation analyses.

From the graphs in Figure 9 (a and b), it was observed that the logarithmic correlations (R^2) indicate moderate explanatory power (0.60 and 0.41, respectively) for H_2S removal and CH_4 increase, respectively, and were much superior to the other tested options. Therefore, in the 0.070 and 0.240 $LO_2.L_{prod.\ biog}^{-1}$ range, increasing the microaeration rate promotes an increase in H_2S removal and CH_4 production gain. However, caution is required not to dilute the biogas

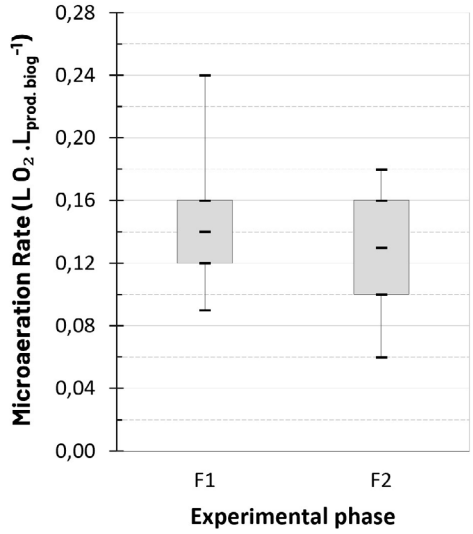


Figure 8 – Microaeration dosages in the experimental phases of the UASB/DC system ($LO_2.L_{biog}^{-1}$).

in methane beyond the useful range for energy recovery. Thus, based on the correlation graphs, it seems more appropriate to work with rates between 0.140 and 0.180 $LO_2.L_{prod.\ biog}^{-1}$ to achieve approximately 40% CH_4 increase and 30% H_2S removal. As mentioned previously, the expected concentrations are around 50% CH_4 and 800 ppm H_2S in the resulting biogas for combustion and energy utilization.

Thus, the most appropriate oxygen dosage, based on the waste gas in relation to the biogas produced by the UASB reactor, would be around 16% (0.140 to 0.180 L $O_2 \cdot L^{-1}$ produced biogas). Compared with the range reported by Krayzelová *et al.* (2015), 0.3–3%, this value appears quite high. However, within the specific context of a UASB reactor treating mainly domestic wastewater, with short residency time in the headspace, it is proved to be the best possible scenario for H_2S removal, from both the biogas and the DC waste gas, while also allowing the recovery of dissolved methane without compromising the biogas through excessive dilution.

Considering this perspective, based on the results and the simplicity and economy of the system, which involves units, equipment, and piping without considerable complexity, it can be concluded that complementing a UASB reactor with a DC system integrating its waste gas to promote microeration inside the three-phase separator constitutes a viable technical and economic option to

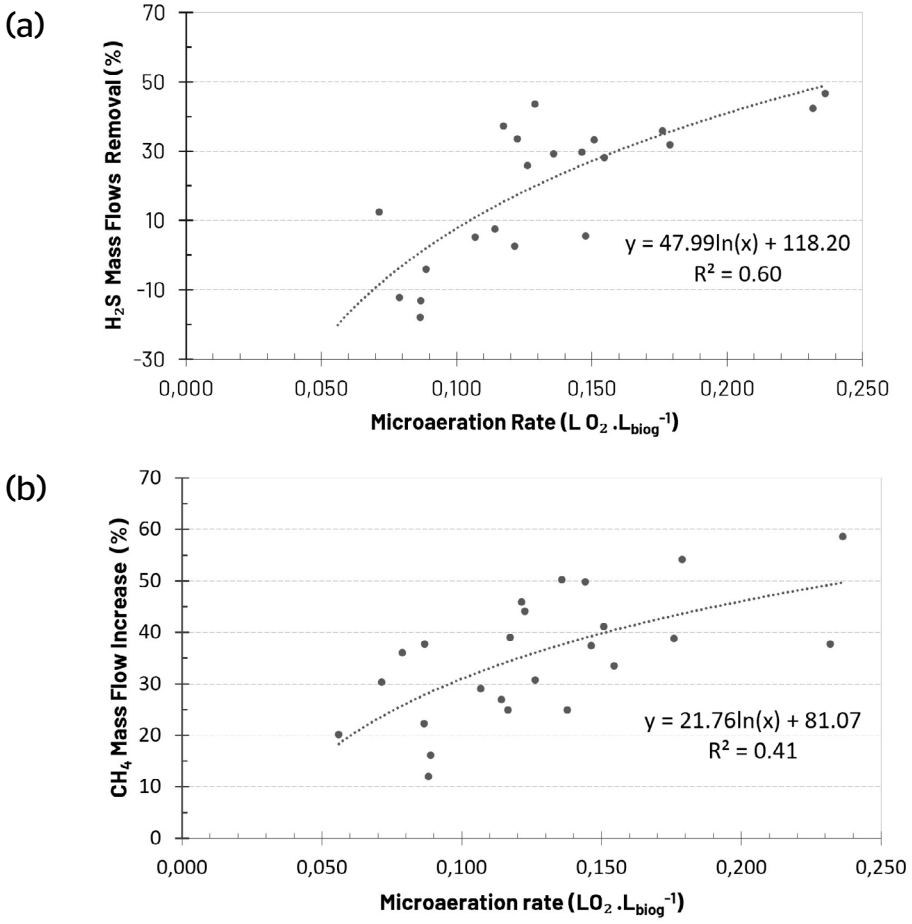


Figure 9 – Relationship between UASB microaeration rate and (a) H_2S mass flows removal and (b) CH_4 mass flow increase in biogas.

be developed, especially when applied in the headspace, recovering dissolved methane along with the biogas.

Oxygen and methane concentrations for operational safety of the UASB/DC system

To discuss safety aspects related to microaerated biogas in the UASB reactor and to waste gas in the DC, the respective graphs (Figures 10a and 10b) show oxygen and methane concentrations over the experimental period (time counted from the onset of microaeration using waste gas, Phase 1). The operating conditions of the gas lines (biogas and waste gas) were: ambient temperature (average around 25°C) and pressure around atmospheric pressure (1 atm); therefore, the safety parameters used below refer to conditions close to these standards, obtained from the flammability diagram between CH_4, O_2, and N_2.

In the microaerated UASB reactor (Figure 10a), O_2 concentrations always remained below 8%, well under the minimum oxygen concentration (MOC) of 12%. This value represents the threshold which combustion cannot occur due to the lack of oxidant/oxygen. Additionally, CH_4 concentrations were much higher than the upper flammability limit (UFL) of 15% in air, a value above which combustion cannot occur due to excess fuel/methane (Crowl, 2023; Guo *et al*., 2024; Mashuga and Crowl, 1998).

For the waste gas in the DC under operation (Figure 10b), O_2 concentrations were expectedly higher, but with median and mean values close to 10%, and thus frequently (over 75% of the data) below the MOC of 12%. O_2 concentrations approached 12% on only three days, and on two additional days they presented values of approximately 13–14%. On the other hand, even when concentrations were close to or slightly above the MOC (12–14%), the minimum CH_4 concentration required for the mixture to be flammable would be within much more restrictive and reduced ranges, approximately 6–10% CH_4 (Crowl, 2003; Mashuga and Crowl, 1998). Methane concentrations in the waste gas were always above 16%, with mean and median values close to 24%.

The 12% MOC value is specifically applied to contexts of methane dilution with nitrogen (N2) and air. When CO2 is present, this limit can rise to 14.5% (Crowl, 2003), making the safety range less restrictive. In the current study, biogas and waste gas had average CO2 concentrations of 6 and 5%, respectively, along with high N2 levels of 35 and 61%, respectively. Due to this composition, the 12% O2 limit is a more conservative and therefore safer approach. Additionally, an extra safety margin is recommended, establishing the operational limit at 10% O2 (Crowl, 2003). Therefore, in future studies and applications, it is recommended to adjust the microaeration rate considering this oxygen limit (10%) in the waste gas as well.

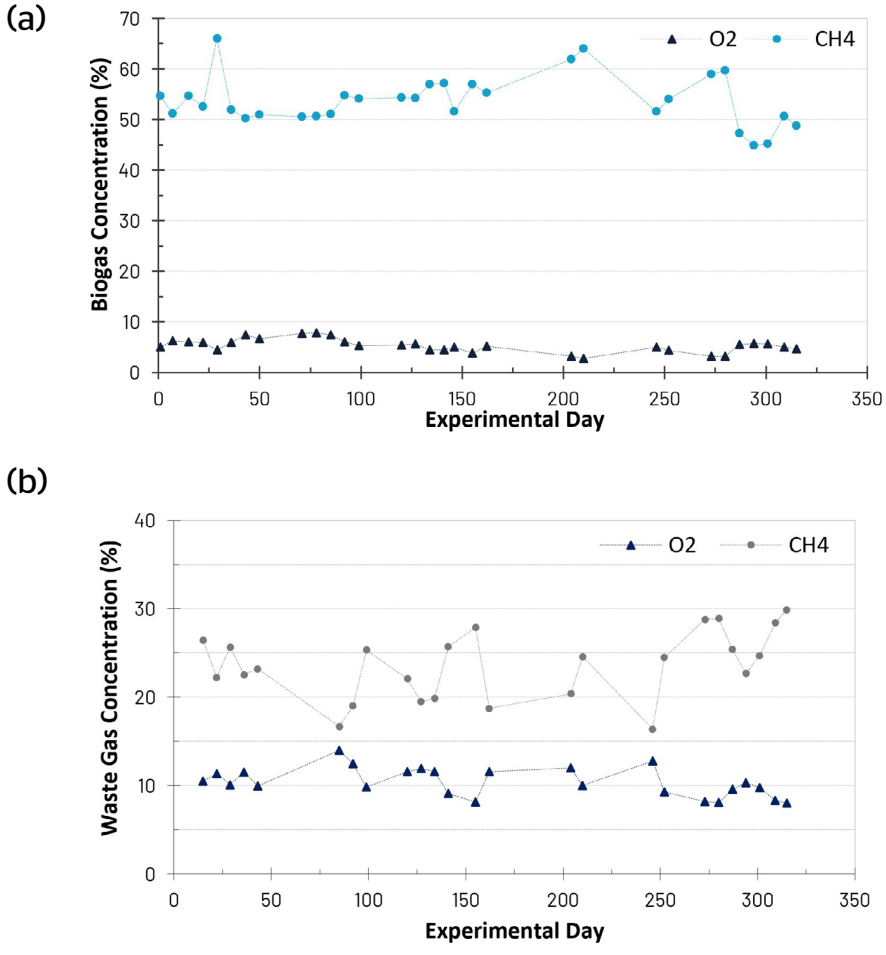


Figure 10 – Time series of O_2 and CH_4 concentration results obtained from the integration of the UASB/DC system: (a) biogas from the microaerated UASB reactor using waste gas, and (b) waste gas from the DC directed to the UASB reactor.

Therefore, under normal operation at these concentration levels, safety risks are considered low, especially for the biogas in the UASB reactor. Under transient operational conditions, it is necessary to plan and implement conscious safety and precautionary measures. Continuous O_2 monitoring in the DC waste gas and occasional application in the UASB biogas are recommended for operational monitoring and control of the system.

CONCLUSION

The results of this study demonstrate that integrating a desorption chamber with microaeration into a UASB reactor is a promising strategy to mitigate dissolved gases, achieving relevant removals of CH_4 and H_2S with low energy demand and simple operation. The use of waste gas as a source of microaeration proved technically feasible. Regarding the injection point, no major differences were observed between the headspace and submerged applications.

This study achieved performance close to, and even exceeding, a 37% increase in CH_4 in the biogas mass flow and a 28% H_2S removal in the mixed biogas and waste gas mass flows. As a characterization of the mixed and treated flow, it was possible to reach concentrations of approximately 50% CH_4 and 800 ppm H_2S, which can be used in various contexts as renewable energy.

As this study was conducted in a demonstration-scale unit treating real domestic wastewater under practical conditions, the findings provide valuable insights into the limitations and the potential of this integrated approach. Nevertheless, challenges remain with oxygen dosing control and the operational variability inherent to full-scale plants, including fluctuations in influent characteristics, flow rate, and biogas production. These results highlight the need for further investigation to establish robust operational guidelines and to advance the scalability of microaeration in anaerobic sewage treatment systems. Future studies should also evaluate alternative control parameters for microaeration, such as the L O_2/L biogas ratio.

ACKNOWLEDGEMENTS

The authors would like to thank the National Council for Scientific and Technological Development – CNPq; the Minas Gerais State Research Support Foundation – FAPEMIG; the Coordination for the Improvement of Higher Education Personnel – CAPES; Pro-Rectory of Research Federal University of Minas Gerais – PRPq/UFMG; the Regulatory Program for Research, Development and Innovation of the Regulatory Agency for Water Supply and Sewage Services of the State of Minas Gerais – ARSAE-MG, and the Minas Gerais Sanitation Company – COPASA; which made this work possible.

AUTHOR'S CONTRIBUTIONS

Mota, D. C.: Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. Batista, I. F.: Formal analysis, Validation, Visualization, Writing – original draft, Writing – review & editing. Souza, C. L.: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

DATA AVAILABILITY STATEMENT

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

REFERENCES

BRESSANI-RIBEIRO, Thiago; CHERNICHARO, Carlos. Upflow anaerobic sludge blanket reactors. *In*: TOMEI, Maria Concetta; GARRIDO, Juan Manuel (Ed.). **Anaerobic Treatment of Domestic Wastewater:** present status and potentialities. Londres: Iwa Publishing, 2024. p. 29-48.

CASTRO, Iacy Maria; AZEVEDO, Lariza; SOUZA, Cláudio Leite de. Evaluation of microaeration strategies in the digestion zone of UASB reactors as an alternative for biogas desulfurization. **Environmental Technology**, v. 44, n. 4, p. 431-445, 2021. https://doi.org/10.1080/09593330.2021.1974574

CHEN, Qing; WU, Wanqing; QI, Dacheng; DING, Yihong; ZHAO, Zihao. Review on microaeration-based anaerobic digestion: state of the art, challenges, and prospectives. **Science of The Total Environment**, v. 710, 136388, 2020. http://doi.org/10.1016/j.scitotenv.2019.136388

CROWL, Daniel. **Understanding Explosions**. New York: American Institute of Chemical Engineers, 2003.

FU, Shanfei; LIAN, Shujuan; ANGELIDAKI, Irini; GUO, Rongbo. Microaeration: an attractive strategy to facilitate anaerobic digestion **Trends in Biotechnology**, v. 41, n. 5, p. 714-726, 2023. http://doi.org/10.1016/j.tibtech.2022.09.008

GLÓRIA, Roberto Meireles; MOTTA, Tomás Miranda; SILVA, Priscila Vieira Oliveira; COSTA, Priscila da; BRANDT, Emanuel Manfred Freire; SOUZA, Cláudio Leite de; CHERNICHARO, Carlos Augusto Lemos. Stripping and dissipation techniques for the removal of dissolved gases from anaerobic effluents. **Brazilian Journal of Chemical Engineering**, v. 33, n. 4, p. 713-721, 2016. https://doi.org/10.1590/0104-6632.20160334s20150291

GUO, Yalou; HU, Guoping; LIU, Liying; WANG, Jian; WEBLEY, Paul; LI, Gang Kevin. Upgrading low-concentration oxygen-bearing coal bed methane by dual-reflux vacuum swing adsorption. **Adsorption**, v. 30, p. 1479-1491, 2024. https://doi.org/10.1007/s10450-024-00513-3

HUETE, Alvaro; DE LOS COBOS-VASCONCELOS, Daniel; GÓMEZ-BORRAZ, Tania; MORGAN-SAGASTUME, Juan Manuel; NOYOLA, Adalberto. Control of dissolved CH_4 in a municipal UASB reactor effluent by means of a desorption - Biofiltration arrangement. **Journal of Environmental Management**, v. 216, p. 383-391, 2018. https://doi.org/10.1016/j.jenvman.2017.06.061

INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE – IPCC. **2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories**. Switzerland: IPCC, 2019.

KRAAKMAN, Norbertus Joannes Richardus; DÍAZ, Israel; FDZ-POLANCO, Maria; MUÑOZ, Raul. Large-scale micro-aerobic digestion studies at municipal water resource recovery facilities for process-integrated biogas desulfurization. **Journal of Water Process Engineering**, v. 53, 103643, 2023. https://doi.org/10.1016/j.jwpe.2023.103643

KRAYZELOVA, Lucie; BARTACEK, Jan; DÍAZ, Israel; JEISON, David; VOLCKE, Eveline; JENICEK, Pavel. Microaeration for hydrogen sulfide removal during anaerobic treatment: A review. **Reviews in Environmental Science and Bio/Technology**, v. 14, n. 4, p. 703-725, 2015. https://doi.org/10.1007/s11157-015-9386-2

MASHUGA, Chad; CROWL, Daniel. Application of the flammability diagram for evaluation of fire and explosion hazards of flammable vapors. **Process Safety Progress**, v. 17, n. 3, p. 176-183 1998. https://doi.org/10.1002/prs.680170305

METCALF, Leonard; EDDY, Harrison. **Wastewater treatment and resource recovery**. 5. ed. Porto Alegre: Amgh, 2016.

PLAS, Christian; HARANT, Hanna; DANNER, Herbert; JELINEK, Eveline; WIMMER, Katharina; HOLUBAR, Peter; BRAUN, Rudolf. Ratio of biological and chemical oxidation during the aerobic elimination of sulphide by colourless sulphur bacteria. **Applied Microbiology and Biotechnology**, v. 36, n. 6, p. 817-822, 1992. https://doi.org/10.1007/BF00172201

SANTO, Belinazir Costa do Espírito; MACHADO, Lucas Martins; MARINHO, Tamile Dafne; SOUZA, Cláudio Leite de. Development of a desorption chamber for controlling dissolved gas emissions from UASB reactor effluent: minimization of bad odors and mitigation of greenhouse effect. *In:* ROSS, Bárbara Zanicotti Leite; POSSETTI, Gustavo Rafael Colleri; GERVASONI, Ronald. (Ed.). **Innovation for sustainability in environmental sanitation:** works awarded in the Inova Sanepar Prize. Curitiba: Sanepar, 2019. p. 16-27.

SOUZA, Cláudio Leite de. **Study of the formation, transport, and consumption pathways of methane and hydrogen sulfide gases**. 2010. 147 p. Doctoral dissertation (Post-Graduation in Sanitary and Environmental Engineering) – Universidade Federal de Minas Gerais, Belo Horizonte, 2010.

SOUZA, Cláudio Leite de; CHERNICHARO, Carlos; AQUINO, Sérgio Francisco. Quantification of dissolved methane in UASB reactors treating domestic wastewater under different operating conditions. **Water Science and Technology**, v. 64, n. 11, p. 2259-2264, 2011. https://doi.org/10.2166/wst.2011.695

SOUZA, Cláudio Leite de; CHERNICHARO, Carlos; MELO, Gilberto. Methane and hydrogen sulfide emissions in UASB reactors treating domestic wastewater. **Water Science e Technology**, v. 65, n. 7, p. 1229-1237, 2012. https://doi.org/10.2166/wst.2012.854

SUN, Qie; LI, Hailong; YAN, Jinying; LIU, Longcheng; YU, Zhizin; YU, Xinhai. Selection of appropriate biogas upgrading technology-a review of biogas cleaning, upgrading and utilisation. **Renewable and Sustainable Energy Reviews**, v. 51, p. 521-532, 2015. https://doi.org/10.1016/j.rser.2015.06.029

WELLINGER, Arthur; LINDBERG, Anna. **Biogas upgrading and utilisation**. Switzerland: IEA Bioenergy, 2001.

© 2026 Associação Brasileira de Engenharia Sanitária e Ambiental
This is an open access article distributed under the terms of the Creative Commons license.